Borsuk-Ulam Theorem

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Introduction & Motivation

> *At any moment, there exist two antipodal points on Earth with* exactly *the same temperature and pressure.*

This is not a coincidence — it is a theorem.

The **Borsuk-Ulam Theorem** (1933) says that any continuous map from a sphere into Euclidean space of one lower dimension must send some pair of antipodal points to the same value.

It was conjectured by Stanislaw Ulam and proved by Karol Borsuk. The result seems geometric but its proof is fundamentally topological — and its consequences reach far beyond geometry.

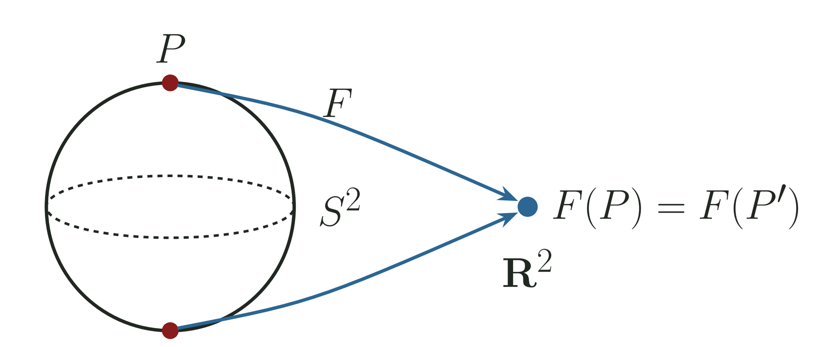
Key Definitions

- $\mathbf{R}$ = real numbers; $\mathbf{C}$ = complex plane; $\mathbf{C}^* = \mathbf{C} \setminus \{0\}$ the punctured plane.
- **Sphere:** $S^2 = \{(x,y,z) \in \mathbf{R}^3 : x^2+y^2+z^2 = 1\}$.
- **Antipodal point:** for P on S^2, P' denotes the antipodal point; $(x,y,z)' = (-x,-y,-z)$.
- **Closed unit disk:** $D = \{z : |z| \le 1\}$, boundary circle $T = \{z : |z| = 1\}$.
- $C(X)$: all continuous $f : X \to \mathbf{C}$.
- $C^*(X) = \{f \in C(X) : f(X) \subset \mathbf{C}^*\}$: functions that never vanish.
- **Proposition 1:** if $f \in C^*(D)$, there exists $h \in C(D)$ with $e^h = f$.

Main Theorem

Theorem (Borsuk-Ulam, 1933). *Let $F : S^2 \to \mathbf{R}^2$ be continuous. Then there exists a pair of antipodal points P and P' such that $F(P) = F(P')$.*

Equivalently: no continuous map $F : S^2 \to \mathbf{R}^2$ can satisfy $F(P) \neq F(P')$ for all $P \in S^2$.



Proof Strategy

Identifying $\mathbf{R}^2$ with $\mathbf{C}$, we regard F as an element of $C(S^2)$. The proof proceeds by **contradiction**.

Step 1 — Assume no antipodal pair maps equally.
Suppose $F(P) \neq F(P')$ for all P in S^2. Define $G : S^2 \to \mathbf{C}$ by $G(P) = F(P) - F(P')$. Then G belongs to $C^*(S^2)$, and $G(P) = -G(P')$ for all P in S^2.

Step 2 — Parametrize via the upper hemisphere.
Define f in $C(D)$ by

$$f(x+iy) = G\left(x,\, y,\, \sqrt{1-x^2-y^2}\right)$$

(identify the upper hemisphere with the closed unit disk). Then $f(z) \neq 0$ for all z in D — i.e., f belongs to $C^*(D)$ — and $f(z) = -f(-z)$ for all z in T.

Step 3 — Apply Proposition 1.
By Proposition 1, there exists h in $C(D)$ such that $f = e^h$.

Step 4 — Derive the contradiction.
Let $k(z) = (1/i\pi)[h(z) - h(-z)]$ for $z \in T$.
For $z \in T$:

$$\exp[i\pi k(z)] = f(z)/f(-z) = -1,$$

so $k(z)$ is an **odd integer**. Since k is continuous and T is connected, k must be constant.
So we have

$$h(1) = h(-1) + ki\pi = h(1) + 2ki\pi,$$

forcing $k = 0$ — impossible since 0 is **not** an odd integer. **Contradiction.** ■

Application: The Ham Sandwich Theorem

Theorem (Stone–Tukey, 1942). *Given any three bounded measurable sets $A_1, A_2, A_3 \subset \mathbf{R}^3$, there exists a single plane that simultaneously bisects all three.*

The sets can be in any configuration — no geometric conditions required.

Proof via Borsuk-Ulam:

Step 1 (Normal vectors). Each point $\mathbf{n} = (n_1, n_2, n_3)$ on the unit sphere S^2 is a unit vector in $\mathbf{R}^3$, i.e. $n_1^2 + n_2^2 + n_3^2 = 1$. We use $\mathbf{n}$ as the *normal direction* of a cutting plane.

Step 2 (The bisecting plane $\Pi(\mathbf{n})$). For each $\mathbf{n} \in S^2$, consider the family of planes with normal $\mathbf{n}$: $\{\mathbf{x} \in \mathbf{R}^3 : \mathbf{n} \cdot \mathbf{x} = t\}$, parameterized by $t \in \mathbf{R}$. By the intermediate value theorem, there is a unique $t(\mathbf{n})$ such that this plane bisects A_1. Define

$$\Pi(\mathbf{n}) \;=\; \{\mathbf{x} \in \mathbf{R}^3 : \mathbf{n} \cdot \mathbf{x} = t(\mathbf{n})\}.$$

Step 3 (Signed measure μ). Let μ denote the Lebesgue measure (volume) in $\mathbf{R}^3$. The plane $\Pi(\mathbf{n})$ splits each A_i into a positive side $A_i^+ = \{\mathbf{x} \in A_i : \mathbf{n} \cdot \mathbf{x} > t(\mathbf{n})\}$ and a negative side $A_i^- = \{\mathbf{x} \in A_i : \mathbf{n} \cdot \mathbf{x} < t(\mathbf{n})\}$.

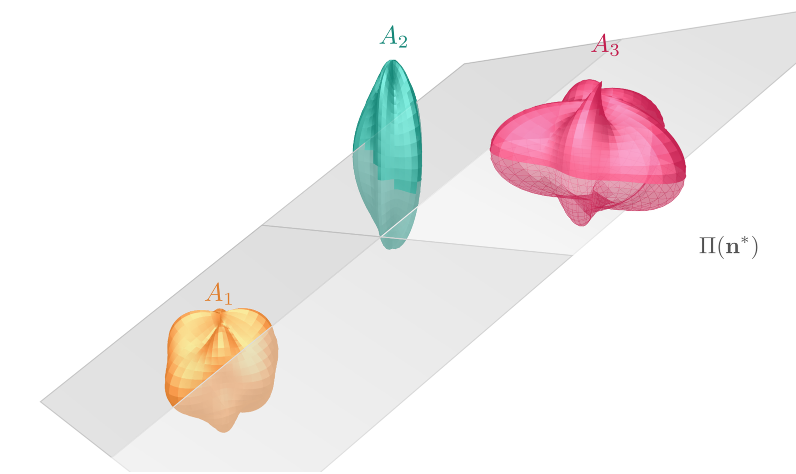
Step 4 (The map F). Define $F : S^2 \to \mathbf{R}^2$ by the volume imbalances of A_2 and A_3:

$$F(\mathbf{n}) \;=\; \left(\mu(A_2^+) - \mu(A_2^-),\; \mu(A_3^+) - \mu(A_3^-)\right).$$

Note F is continuous. Flipping $\mathbf{n} \to -\mathbf{n}$ swaps the positive and negative sides, so $F(-\mathbf{n}) = -F(\mathbf{n})$.

Step 5 (Apply Borsuk-Ulam). By the Borsuk-Ulam theorem, $\exists\, \mathbf{n}^*$ with $F(\mathbf{n}^*) = F(-\mathbf{n}^*) = -F(\mathbf{n}^*)$, so $F(\mathbf{n}^*) = (0,0)$.

Thus $\Pi(\mathbf{n}^*)$ bisects A_2 and A_3 as well, while already bisecting A_1 by construction. ■



Conclusions & Further Directions

This poster proves the $S^2 \to \mathbf{R}^2$ case of the Borsuk-Ulam theorem. Future work could explore the generalized theorem: every continuous map $F : S^n \to \mathbf{R}^n$ identifies some antipodal pair. Additionally, one may investigate the generalized Ham Sandwich theorem, which states that any n measurable objects in $\mathbf{R}^n$ can be simultaneously bisected by a single $(n-1)$-dimensional hyperplane.

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